Data Analyst

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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer

I'll start by loading and exploring the dataset to understand its structure and key characteristics. Then, I'll be ready to address any specific analytical questions you have.

Let me first inspect the dataset.

Key Characteristics of the Dataset:

- **Size**: 231,636 rows × 19 columns.
- **Data Types**:
  - 12 integer columns (e.g., `StateCode`, `MunicipalityCode`, `MonthIssued`).
  - 7 floating-point columns (e.g., `FundingValue`, `InvestmentValue`, `ContractTotalValue`).
- **Columns of Interest**:
  - `ContractTotalValue`: Represents the total contract value, likely a key metric.
  - `FundingValue`, `InvestmentValue`, `CommercializationValue`, `IndustrializationValue`: Different financial aspects of municipal contracts.
  - `EconomicActivity`, `ProgramCode`, `ResourceSourceCode`: Categorization fields.

Now, what specific analytical question do you want to address using this dataset? [>-]

Does the state contribute to an increase in the total contract value?

To determine whether the **state (StateCode)** contributes to an increase in the **total contract value (ContractTotalValue)**, I will:

1. **Analyze the distribution** of contract values by state.
2. **Check for variations** across states—do some states consistently have higher contract values?
3. **Perform a statistical test or regression** to evaluate if the state has a significant impact on contract value.

Let's start by summarizing the contract values per state.

Observations:

- **Total Contract Value Varies by State**:
  - The top state (`StateCode = 19`) has a total contract value of **18.9 billion**.
  - The lowest (`StateCode = 4`) has **16.1 million**—a massive disparity.
- **Average Contract Value Also Varies**:
  - `StateCode = 14` has the highest average contract value (**1.96 million per contract**).
  - `StateCode = 16` has one of the lowest (**146k per contract**).

This suggests that **state plays a role in contract value variations**. I will now check the statistical significance of this effect using an **ANOVA test**, which determines whether the differences in contract values across states are statistically significant. [>-]

Statistical Significance:

- **F-statistic = 38.58, p-value ≈ 2.42×10^{-194}**.
- Since the p-value is **extremely small (≪ 0.05)**, we reject the null hypothesis.
- This confirms that **StateCode has a statistically significant effect on ContractTotalValue**.

Conclusion:

Yes, the **state contributes to an increase in total contract value**, as states significantly impact the distribution of contract values.

Would you like a deeper analysis, such as a regression model to quantify the effect of each state? [>_]

Message Data Analyst